

Hybrid Automated Fact-Checking Pipeline for Climate Claim Verification

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Abstract

Climate misinformation is a growing problem in the world and its spread makes manual fact checking almost impossible. This paper presents an automated fact checking system for climate claim verification built as a two stage pipeline that first retrieves relevant evidence from a corpus of over 1.2 million passages and then classifies each claim as SUPPORTS, REFUTES, NOT_ENOUGH_INFO, or DISPUTED. For retrieval I combined four signals: BM25, dense retrieval using all-mpnet-base-v2 with FAISS, TF-IDF, and query expanded BM25 with Reciprocal Rank Fusion and reranked using a cross encoder. For classification I compared a BiLSTM, BERT-base, and RoBERTa-base, all trained with label smoothing and inverse-frequency class weighting to handle the dataset’s class imbalance. RoBERTa-base achieved the best dev accuracy at 0.7143 on gold evidence, and the full end-to-end pipeline reached a harmonic mean of 0.2525. I also compared my approach against a separate system that uses DeBERTa-v3-large with LoRA and a top- K evidence selection strategy, which achieved a harmonic mean of 0.3186, and used it to analyse the trade-offs between model scale, parameter efficiency, and retrieval design.

1 Introduction

Climate change is one of the most well documented phenomena yet misinformation about it spreads like wildfire. This misinformation can many times negatively change public opinion and influence policy. Automated fact checking systems are one way to address this at scale.

The task here is that given a claim, the system has to find the most relevant evidence passages from a large corpus and then decide whether the claim is supported, refuted, disputed, or just doesn’t have enough information to make a call. Both of these parts are equally tricky because they’re

not independent as a bad retrieval can make the classification worse and a weak classifier can fail even when the retrieval is perfect.

In this dataset the corpus has over 1.2 million evidence passages, so exact keyword matching misses a lot and semantic search has to cover for it. Climate-specific terms like CO₂ or “global mean temperature” appear constantly across many documents, which makes it hard to distinguish relevant evidence from just topically related text. The training set is also relatively small at 1228 claims, and the label distribution is extremely skewed: DISPUTED only has 124 examples compared to 519 for SUPPORTS, which causes the model to ignore some classes if you don’t account for it properly.

To deal with these challenges I built a hybrid retrieval system that combines sparse and dense signals and fuses them with Reciprocal Rank Fusion with cross encoder reranking to get the best candidates. For classification I experimented with a BiLSTM, BERT-base, and RoBERTa-base and used label smoothing with inverse frequency class weights across all three to handle the imbalance. I also attempted DeBERTa-v3-small and DeBERTa-v3-large with LoRA before settling on RoBERTa-base as my primary model. For context and comparison I reference a separate system built by another group that took a different approach to both retrieval and classification, and compared where each design makes a different trade-off.

2 Approach

2.1 Baseline

Before building anything complex I started with a simple claim only baseline using TF-IDF with Logistic Regression. There was no evidence retrieval, just the claim text vectorised and fed into a linear classifier. It got 0.4675 accuracy on the dev set which isn’t terrible considering it has no access to evidence at all, but it also has no real un-

understanding of what the claim is saying, just surface level keyword patterns. This perfectly explains the class imbalance we are dealing with as SUPPORTS makes up $\sim 42\%$ of the training set and so guessing SUPPORTS every time will also get you a $\sim 42\%$ accuracy.

2.2 Evidence Retrieval

The retrieval stage is the first and the hardest part of this task. With 1.2 million passages the system can't just look at everything each time, it needs to get a small set of candidates that are actually relevant. I tested multiple retrieval systems and eventually ended up building four separate retrieval signals and combined them.

2.2.1 BM25

BM25 (Robertson et al., 1994) is a sparse lexical retriever and it scores documents based on term frequency and document length normalisation. It's fast and works well when the claim and the evidence share the same words but it misses cases where different words are used but the meaning is the same. BM25 was used as one of the four signals and everything was tokenised to lowercase with punctuation removed.

2.2.2 Dense Retrieval

To catch semantic cases that BM25 would have missed, I used all-mpnet-base-v2 (Reimers and Gurevych, 2019), a sentence transformer that encodes text into 768 dimensional embeddings. All 1.2 million evidence passages were encoded and stored in a FAISS IndexFlatIP index (Johnson et al., 2021) for fast inner product search. At query time the claim gets encoded and the top 100 nearest passages are retrieved. I had to use chunked blocks of 10,000 with checkpoint saving to encode the full corpus as it takes a while to encode and save all.

2.2.3 TF-IDF

I also included TF-IDF as a third signal using sklearn's TfidfVectorizer with up to 100,000 features, bigrams, and sublinear TF scaling. At query time the claim gets vectorised and cosine similarity is computed against the full sparse matrix to get the top 100 candidates. It came out as the weakest of the three individual retrievers however in the fusion setup it still contributed to improving the overall retrieval score.

2.2.4 Query Expansion

One thing I noticed is that claims are often phrased very differently from the evidence that supports them. To help with this I added a query expansion step where the top 3 BM25 results for the original claim are retrieved, and the 5 most frequent non-stopword terms in those results that aren't already in the claim are added to the query. The expanded query is then run through BM25 again for another set of 100 candidates. This gave a fourth signal without needing any extra models.

2.2.5 Hybrid Fusion with RRF

Now that I had my four signals I combined them using Reciprocal Rank Fusion (Cormack et al., 2009) with $k = 60$, which scores each document based on its rank across all four lists using the formula $\text{score}(d) = \sum 1/(k + \text{rank})$. The main advantage of RRF is that it doesn't care about the actual scores from each retriever, just the ranks, so there's no need to normalise across different scoring scales. The top 100 from the fused list go to the reranker.

2.2.6 Cross-Encoder Reranking

The cross-encoder/ms-marco-MiniLM-L-6-v2 (Nguyen et al., 2016) takes the claim and each candidate evidence passage together as a single input and scores how relevant they are to each other. This is much more accurate than bi-encoder similarity because the model can attend to both texts at once, but it's also much slower which is why I only ran it on the top 25 candidates from RRF. The top 5 from the reranker become the final retrieved evidence for each claim.

2.3 Claim Classification

After the evidence is retrieved, classification is the next step. All three models used label smoothing (Müller et al., 2019) with $\varepsilon = 0.1$ and inverse-frequency class weights to deal with the class imbalance. The class weights came from the training label distribution: SUPPORTS=0.592, REFUTES=1.543, NOT_ENOUGH_INFO=0.795, DISPUTED=2.476. DISPUTED gets weighted over $4\times$ more than SUPPORTS because it only has 124 training examples versus 519.

The transformer models BERT and RoBERTa take the claim concatenated with up to three evidence passages separated by [SEP] tokens, truncated to 256 tokens, and were trained with AdamW (Loshchilov and Hutter, 2019), a linear warmup scheduler over the first 10% of steps, and early

stopping at patience=3. The BiLSTM works differently and is described in its own section below.

2.3.1 BiLSTM

I implemented a BiLSTM as the sequence modelling component. Rather than training embeddings from scratch, I used all-mpnet-base-v2 to encode each input into a 768 dimensional vector, then reshaped it into a 12×64 sequence of timesteps to feed into the LSTM. The model is two bidirectional LSTM layers with hidden size 256, with attention pooling over the hidden states and a two layer MLP classifier with GELU activation and dropout. It was trained with AdamW at $\text{lr}=5\text{e-}4$ and early stopping at patience=5.

2.3.2 BERT-base

I fine-tuned bert-base-uncased (Devlin et al., 2019) by taking its [CLS] token representation and projecting it to four output classes. Trained with $\text{lr}=1\text{e-}5$, weight decay=0.05, label smoothing, and the class weights described above. Early stopping with patience=3 over a max of 15 epochs.

2.3.3 RoBERTa-base

I fine-tuned RoBERTa-base (Liu et al., 2019) under the same configuration as BERT. The main reason to try it was that RoBERTa’s pretraining is more thorough — it uses more data, dynamic masking, and drops the next sentence prediction objective that BERT uses. Same $\text{lr}=1\text{e-}5$, weight decay=0.05, max 15 epochs, patience=3.

2.3.4 Attempted: DeBERTa with LoRA

I also attempted both DeBERTa-v3-small and DeBERTa-v3-large with LoRA as parameter efficient alternatives. DeBERTa-v3-small reached 0.52 dev accuracy and DeBERTa-v3-large with LoRA reached 0.55, both trained for 10 epochs with the same $\text{lr}=1\text{e-}5$ and early stopping configuration as BERT and RoBERTa. Neither improved over RoBERTa-base so I focused on RoBERTa for the remainder of the experiments.

2.4 Final Model

Once RoBERTa was selected as the primary model I retrained it on the combined train and dev set, 1382 samples total, for 12 epochs with no early stopping, since there is no held-out validation set at this stage and the goal is to maximise exposure to all available labelled data. Everything else stayed the same.

2.5 Comparison System

For context I compare my results against a system built by my old group (Group 36). Their retrieval pipeline uses BM25 with Porter stemming and stopword removal combined with all-MiniLM-L6-v2 for dense retrieval, fused with RRF over a top 300 candidate pool, then reranked with the same cross-encoder. They also added a margin-gated dynamic top- K selection step after reranking, where instead of always picking a fixed top 5, the number of evidence passages selected per claim is determined by the score margin between candidates. Their classifier is DeBERTa-v3-large fine-tuned with LoRA, training only 0.36% of the model’s parameters. Their final pipeline achieved $F=0.2229$, $A=0.5584$, $H=0.3186$.

3 Experiments

3.1 Dataset

The dataset has three splits: 1228 training claims, 154 dev claims, and 153 test claims, with a corpus of 1,208,827 evidence passages to retrieve from. The label distribution in training is heavily skewed toward SUPPORTS with 519 examples, then NOT_ENOUGH_INFO with 386, REFUTES with 199, and DISPUTED with just 124. This imbalance is what led to the class weightings described in Section 2.

3.2 Evaluation

Three metrics are used. Evidence Retrieval F-score measures how well the retrieved passages match the gold evidence, computed per claim and averaged. Claim Classification Accuracy is standard accuracy on the predicted label. The Harmonic Mean $H = 2FA/(F + A)$ is the main metric and is designed so that a system can’t just optimise one component and ignore the other.

Classification accuracy is also evaluated separately on gold evidence, where the correct evidence is given directly to the classifier rather than using what the retriever found. This gives an upper bound on what the classifier can do and separates retrieval errors from classification errors.

3.3 Retrieval Results

Table 1 shows the retrieval F-score for each individual retriever and the full hybrid system evaluated on the dev set.

Dense retrieval outperforms BM25 on its own, which makes sense since the climate domain has a

Retriever	F-score
TF-IDF only	0.0569
BM25 only	0.1053
Dense only (all-mpnet-base-v2)	0.1507
Hybrid + RRF + cross-encoder	0.1850

Table 1: Retrieval F-score on dev set (top-5)

lot of paraphrasing and semantic variation that keyword matching alone can't handle. TF-IDF is the weakest of the three and even though it is similar to BM25 on the surface, it lacks the term saturation and length normalisation that BM25 has. The hybrid system beats all three individual retrievers, confirming that the four signals are complementary rather than redundant.

Table 2 shows Recall@K for the hybrid system, which gives a better picture of how many claims have at least one correct evidence passage in the top K results.

K	Recall
5	0.3896
10	0.4610
20	0.5584
50	0.6818

Table 2: Hybrid Recall@K on dev set

At $K=50$ the retriever finds at least one correct passage for 68% of claims, but at $K=5$ that drops to 39%. The gap between Recall@5 and the retrieval F-score shows that the system is finding relevant evidence but not always ranking it high enough, which is the problem the cross-encoder reranker was trying to fix.

3.4 Classification Results

Table 3 shows the dev accuracy for each classifier evaluated on gold evidence.

Model	Dev Accuracy
BiLSTM	0.4416
BERT-base	0.5584
RoBERTa-base	0.7143

Table 3: Classifier comparison on dev set (gold evidence)

RoBERTa improves significantly over BERT despite being a similar architecture trained under the same conditions, which lines up with what you'd expect given RoBERTa's stronger pretraining. The BiLSTM sits at 0.4416, close to the claim-only

baseline of 0.4675, suggesting it struggled to extract useful signal from the evidence embeddings in the way it was set up.

Table 4 shows the per-class breakdown for RoBERTa on gold evidence.

Class	Precision	Recall	F1
SUPPORTS	0.7317	0.8824	0.8000
NOT_ENOUGH_INFO	0.7692	0.7317	0.7500
REFUTES	0.6667	0.5185	0.5833
DISPUTED	0.5000	0.3333	0.4000

Table 4: RoBERTa-base per-class results (gold evidence)

SUPPORTS and NOT_ENOUGH_INFO are the strongest at $F1=0.80$ and 0.75 respectively. DISPUTED is the weakest at $F1=0.40$ despite the class weighting, which makes sense given it only has 18 examples in the dev set and the label itself is inherently ambiguous, a disputed claim sits somewhere between supported and refuted which makes it genuinely hard to classify.

3.5 Full Pipeline Results

Table 5 shows the end-to-end results when the retriever and classifier run together.

Metric	Value
Evidence Retrieval F-score (F)	0.1682
Claim Classification Accuracy (A)	0.5065
Harmonic Mean (H)	0.2525

Table 5: Full pipeline results on dev set

The drop from 0.7143 gold evidence accuracy to 0.5065 pipeline accuracy shows how much retrieval errors hurt classification when wrong evidence is retrieved the classifier works with bad inputs and there's only so much it can do.

3.6 Ablation Study

Table 6 shows what happens to the full pipeline metrics when each retriever is used alone instead of the hybrid, keeping RoBERTa-base as the fixed classifier.

Retriever	F	A	H
TF-IDF only	0.0569	0.4545	0.1012
BM25 only	0.1053	0.4740	0.1723
Dense only	0.1507	0.5000	0.2316
Hybrid + RRF + rerank	0.1682	0.5065	0.2525

Table 6: Retriever ablation (RoBERTa-base classifier)

The hybrid does better on all three metrics and better retrieval F-score generally lifts classification accuracy too, not just the harmonic mean.

3.7 Comparison with Alternative System

Table 7 compares my pipeline against my old team’s system described in Section 2.5.

System	F	A	H
RoBERTa-base (mine)	0.1682	0.5065	0.2525
DeBERTa-v3-large + LoRA	0.2229	0.5584	0.3186

Table 7: Pipeline comparison

The DeBERTa system has a higher H mainly because of its retrieval F-score 0.2229 vs 0.1682 is a meaningful gap. Their BM25 uses Porter stemming which helps with morphological variants and their margin-gated dynamic top- K means they’re selecting a variable number of evidence passages per claim rather than always taking 5, both of which improve retrieval precision. On classification the difference is smaller: their A is 0.5584 vs my 0.5065, and given that my RoBERTa-base gets 0.7143 on gold evidence versus their best reported classifier at 0.5584, the gap in pipeline accuracy is mostly explained by retrieval quality rather than classifier quality.

4 Analysis

4.1 Retrieval Analysis

From the retrieval results it’s clear that no single method is enough on its own. BM25 is fast and decent but misses anything phrased differently from the claim. Dense retrieval is more reliable but slow and still misses a lot. TF-IDF on its own was basically useless, but together the three plus query expansion created a much better hybrid that achieved the best F-score, as the four signals fail in different areas and cover for each other.

The Recall@K numbers tell a better story than just the F-score alone. At K=50 the retriever finds at least one correct passage for 68% of claims but the F-score is only 0.1682. This gap exists because F-score penalises you for retrieving any wrong evidence alongside the correct one, if you get 1 correct out of 5 your precision is only 0.2, dragging the average down. So the retriever is finding relevant evidence for most claims but not ranking them cleanly at the top 5.

The comparison system’s retrieval F of 0.2229 vs my 0.1682 comes down to Porter stemming

on BM25, which maps “warming”, “warm”, and “warmed” all to the same root term — really useful in a climate domain, and their margin-gated dynamic top- K which reduces the noise that a fixed top-5 introduces.

4.2 Classifier Analysis

The jump from BiLSTM at 0.4416 to BERT at 0.5584 to RoBERTa at 0.7143 on gold evidence is clear. The BiLSTM hit 0.4416 on its first epoch and never improved, simply plateauing for 5 epochs before early stopping triggered. This suggests the model got stuck close to the majority class distribution and couldn’t extract useful signal from the evidence embeddings in the way they were structured. Reshaping a flat 768 dimensional embedding into a 12×64 sequence and treating those chunks as timesteps was a forced setup because the LSTM had no real reason to find meaningful patterns across the chunks.

RoBERTa’s per-class results show SUPPORTS and NOT_ENOUGH_INFO are the strongest at F1=0.80 and 0.75 respectively. REFUTES and DISPUTED drop to F1=0.58 and 0.40, which makes sense since DISPUTED is inherently ambiguous, sits somewhere between supported and refuted, and only has 18 dev examples. The class weight of 2.476 helps but isn’t enough to fully compensate.

The drop from gold evidence accuracy of 0.7143 to pipeline accuracy of 0.5065 shows how much retrieval quality matters. Even with a strong classifier, wrong evidence leads to wrong predictions.

4.3 Error Analysis

76 out of 154 dev claims were misclassified in the full pipeline, a 49.4% error rate. The most common error patterns were SUPPORTS predicted as NOT_ENOUGH_INFO (16 cases) and NOT_ENOUGH_INFO predicted as SUPPORTS (14 cases). These are mirror errors of each other and the boundary between the two labels is genuinely blurry, especially when the retrieved evidence is partially relevant but not conclusive.

DISPUTED was the biggest problem. Out of 18 DISPUTED claims only 4 were correctly predicted and 13 were classified as SUPPORTS. With only 124 training examples the model doesn’t have enough to really learn what disputed means in this context.

The retrieval vs classification breakdown is also telling:

- Retrieval hit + Classification correct: 44

- Retrieval hit + Classification wrong: 29
- Retrieval miss + Classification correct: 34
- Retrieval miss + Classification wrong: 47

Even when the retriever finds the right evidence the classifier gets it wrong 29 out of 73 times ($\sim 40\%$). It's not just a retrieval problem. And 34 claims were classified correctly despite a retrieval miss, which tells us the two components aren't as tightly coupled as you might expect.

I looked at this specific case: "The world is now 1C warmer than pre-industrial times" — the retriever got the exact gold evidence, both passages matched perfectly, but the classifier still predicted NOT_ENOUGH_INFO. The evidence clearly supports the claim but the model treated it as ambiguous, which points to a classifier confidence issue rather than anything wrong with retrieval.

4.4 Strengths and Limitations

The main strength of the system is the classifier. 0.7143 on gold evidence is strong for a 4-class problem on a small training set and the label smoothing and class weighting setup clearly helped given how imbalanced the data was.

The main limitation is retrieval. $F=0.1682$ is the bottleneck and most of the gap between my harmonic mean and the comparison system's comes from here. Adding stemming to BM25 and using a smarter evidence selection strategy would likely push retrieval F up meaningfully.

Another limitation is the small training set. 1228 claims is not a lot to learn from, especially for DISPUTED which only has 124 examples. More training data would help the classifier generalise better, particularly on minority classes where even with class weighting the model struggled.

5 Conclusion

The system I built works well on classification but retrieval is the clear bottleneck. RoBERTa-base with label smoothing and inverse-frequency class weighting achieved 0.7143 accuracy on gold evidence, which is strong for a 4-class problem on a small and imbalanced dataset. The full pipeline drops to 0.5065 classification accuracy and a harmonic mean of 0.2525, and that gap is almost entirely explained by retrieval quality.

The hybrid retrieval system combining BM25, dense retrieval, TF-IDF, and query-expanded BM25 with RRF and cross-encoder reranking does improve over any individual retriever, but $F=0.1682$

still leaves a lot of claims without the right evidence going into classification.

For future work, improving the retrieval side would have the biggest impact. Adding stemming to BM25, using larger rerank pools, and looking into smarter evidence selection strategies would likely push the harmonic mean up more than any further classifier tuning would.

6 Team Contributions

Manav Gupta: built the full retrieval pipeline including BM25, dense retrieval with all-mpnet-base-v2 and FAISS, TF-IDF retrieval, query expansion, RRF fusion, and cross-encoder reranking. Implemented all three classifiers, BiLSTM, BERT-base, and RoBERTa-base, with label smoothing and inverse-frequency class weighting. Managed all checkpointing and memory optimisation throughout.

Group 36: results from my old group are referenced in this report as a comparison and contextual reference. None of their code or implementations were used in my own system.

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